

SECTION 26 05 00

COMMON WORK RESULTS FOR ELECTRICAL

Riverside Mixed-Use Development
Project No. RMD-2025-001

PART 1 - GENERAL

1.01 REFERENCES

- NFPA 70 - National Electrical Code (NEC) 2023 Edition
- NFPA 72 - National Fire Alarm and Signaling Code
- IEEE C2 - National Electrical Safety Code
- UL 67 - Panelboards
- UL 891 - Switchboards
- IEEE 1584 - Arc-Flash Hazard Calculations
- NFPA 70E - Standard for Electrical Safety in the Workplace

1.02 SUBMITTALS

- A. Shop drawings for switchboards, panelboards, and motor control centers.
- B. Load calculations for each panel and switchboard.
- C. Short circuit and coordination study.
- D. Arc flash hazard analysis per IEEE 1584 and NFPA 70E.
- E. Equipment cut sheets with catalog numbers.

1.03 QUALITY ASSURANCE

- A. Electrical contractor: licensed in Commonwealth of Virginia.
- B. All equipment UL listed or labeled.
- C. Arc flash labels per NFPA 70E on all equipment rated 50V or more.

PART 2 - PRODUCTS

2.01 POWER DISTRIBUTION

Equipment	Rating	Voltage	Config	Manufacturer
Main Switchboard	2000A	480/277V	3PH 4W	Square D / Eaton / Siemens
Distribution Panel	400A	480/277V	Bolt-on	Square D / Eaton / Siemens
Branch Panel	225A	208/120V	Bolt-on	Square D / Eaton / Siemens
Transformer	150 kVA	480-208/120V	Dry Type	Square D / Eaton / Siemens

2.02 CONDUCTORS

Application	Min Size	Material	Insulation	Voltage
Branch circuits	#12 AWG	Copper	THHN/THWN-2	600V
20A over 100 ft	#10 AWG	Copper	THHN/THWN-2	600V
Feeders <= 1/0	Per calc	Copper	THHN/THWN-2	600V
Feeders > 1/0	Per calc	Aluminum	XHHW-2	600V

Color coding: black/red/blue phases, white neutral, green ground.

2.03 RACEWAYS

- A. EMT: ANSI C80.3, for concealed dry locations.
- B. IMC: ANSI C80.6, for exposed and outdoor locations.
- C. Rigid PVC: Schedule 40 underground, Schedule 80 exposed.
- D. Cable tray: aluminum, ladder type, 12-inch minimum width.
- E. Minimum conduit size: 3/4-inch trade size.

2.04 LIGHTING

- A. LED throughout, minimum DLC Premium listed.
- B. Office: recessed 2x4 troffer, 4000K, 40+ lumens/watt, 0-10V dimming.
- C. Corridors: recessed linear, 3500K, integral emergency battery.
- D. Parking: high-bay LED, 5000K, IP65, occupancy sensor.

2.05 FIRE ALARM

- A. Addressable fire alarm system per NFPA 72.
- B. FACP: Notifier NFS2-3030 or approved equal.
- C. Smoke detectors: photoelectric, addressable, integral sounder base.
- D. Pull stations: double-action, addressable, at all required exits.
- E. Notification: horn/strobe, wall-mounted, candela per room size.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. Install per NEC and manufacturer's instructions.
- B. All splices in accessible junction boxes; no splices in raceways.
- C. Conductor fill: maximum 40% of conduit area per NEC Chapter 9.
- D. Minimum 12-inch separation between power and low-voltage raceways.

3.02 GROUNDING

- A. Equipment grounding: copper, sized per NEC Table 250.122.
- B. Ground bus in each panelboard and switchboard.
- C. Grounding electrode system per NEC Article 250.
- D. Lightning protection: UL 96A master-labeled system.

3.03 TESTING AND COMMISSIONING

- A. Megger test all feeders: minimum 100 megohms at 1000V DC.
- B. Ground resistance: maximum 5 ohms per electrode.
- C. Circuit continuity test on 100% of branch circuits.
- D. Functional test of all fire alarm devices and sequences.
- E. Phase rotation verification on all 3-phase equipment.
- F. Thermal scan under load of all panelboard and switchboard connections.

END OF SECTION 26 05 00